

tenXengage

Redemption Store Integration

Business Requirements Document

Cash & Non-Cash Reward Redemption · XTRM & Xoxoday Integration · Dual Wallet · Ledger Engine

Platform: tenXengage

Feature Area: Rewards & Redemption

Vendors: XTRM (Cash) · Xoxoday (Non-Cash)

Tiers: Starter · Professional · Enterprise

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Highly Confidential

1. Document Purpose

This BRD defines the requirements for the Redemption Store integration within the tenXengage AI-Native Partner Activation Network. It covers the complete reward redemption lifecycle — from wallet balance management and currency-aware catalog display through to vendor fulfillment, ledger finalization, returns, and audit.

The document is structured to be directly usable for implementation planning, schema design, API contracts, and phased execution. It governs two concurrent integration tracks:

- XTRM — cash-based payouts including ACH, PayPal, Venmo, and international bank transfers across 180+ countries
- Xoxoday — non-cash reward catalog covering merchandise, gift cards, prepaid cards, travel experiences, and vendor-sponsored events

TenXengage owns the reward wallet ledger, the end-to-end user experience, and all balance accounting. Both vendors operate as fulfillment engines only — the user sees a single unified Redemption Store with no exposure to the underlying vendor routing.

2. Product Vision

Close the partner reward loop by giving every partner seller and partner company a seamless, currency-aware redemption experience that converts earned rewards into real, tangible value — without friction, without compliance burden, and without exposing internal infrastructure.

Positioning	Description
Platform	tenXengage AI-Native Partner Activation Network
Feature Module	Redemption Store Integration
Cash Payout Vendor	XTRM — global cash disbursement platform
Non-Cash Vendor	Xoxoday — global rewards and gift catalog
Ledger Owner	TenXengage (authoritative balance and transaction record)
Alternative Labels	Earn-to-Redeem Engine · Partner Rewards Wallet · Digital Rewards Store

3. Why This Matters

Partners earn rewards across every touchpoint on tenXengage — incentive programs, training completions, activity challenges, journey milestones, community contributions, and deal closures. Today, reward accumulation is tracked but no redemption mechanism exists. Partners have no way to convert their earned balances into tangible value, which directly undermines engagement, platform stickiness, and program ROI.

The Redemption Store closes this loop. It makes the platform's earn-and-redeem promise real:

- Partner sellers are more motivated to participate in programs when earned rewards are immediately redeemable
- Vendors (clients) gain a powerful retention and activation lever — real rewards tied to real behavior
- The dual-wallet model (individual + company) supports both personal and team-level incentive structures

- The ledger-first architecture ensures every balance movement is auditable, reversible on failure, and independent of vendor uptime
- XTRM and Xoxoday together cover the full reward spectrum — from global cash payouts to localized non-cash catalogs — without tenXengage building fulfillment infrastructure

4. Current-State Foundation

The existing platform already includes the following, which this feature preserves and extends:

- Reward transactions tracked via the existing TransactionStatus and TransactionType enums (REWARD, REDEMPTION, ADJUSTMENT, BONUS, CLAWBACK, TRANSFER)
- Four defined currency types: cash, points, credits, tickets — no new types are introduced
- Incentive programs (SALES, TRAINING, ACTIVITY, JOURNEY) that already generate reward earning events
- Compliance module with anti-bribery controls, government deal restrictions, value cap monitoring, and ComplianceAlertType framework
- Audit logging via @Audited annotation with AuditAction and AuditResourceType enums
- Notification framework with NotificationCategory.REWARDS already defined
- Permission system (5-layer resolution) and feature flag infrastructure (FeatureFlagService, tier-based enablement)
- Multi-tenant isolation via client_id, Hibernate @Filter, TenantAware interface, and TenantEntityListener

5. Product Goals

5.1 Business Goals

- Activate the earn-to-redeem loop — make every earned reward convertible into real value
- Increase partner engagement and program participation by making rewards tangible and immediately accessible
- Give Client Admins a configurable, tenant-specific redemption catalog they can control without platform team involvement
- Support global partner ecosystems with international cash payouts (XTRM) and localized non-cash catalogs (Xoxoday)
- Maintain a clean compliance posture by delegating KYC/AML and tax reporting to XTRM for all cash disbursements
- Provide a full, exportable audit trail for all redemption activity at the tenant and user level

5.2 Product Goals

- Deliver a unified Redemption Store UI that is currency-aware, catalog-filtered, and vendor-agnostic from the user's perspective
- Implement a robust, two-step ledger model: reserve on submission, finalize debit on vendor confirmation
- Support dual wallet structures: individual seller wallets and pooled partner company wallets
- Implement a complete returns flow for non-cash redemptions: user-initiated, Client Admin approval-gated, Xoxoday confirmation required
- Expose clean APIs for wallet balances, redemption catalog, transaction history, return management, and vendor webhooks
- Ensure every ledger mutation is written as an immutable LedgerEntry before wallet totals are updated

5.3 Non-Goals for Initial v1

- Building a native redemption fulfillment engine or merchandise catalog — delegated to XTRM and Xoxoday
- Supporting currency types beyond the four platform-defined types: cash, points, credits, tickets
- Cross-currency redemptions (e.g., combining points and tickets in a single transaction)
- Partial returns — only full-amount returns are supported in v1
- Real-time currency conversion between reward types
- Native mobile redemption experience — initial release targets the web platform
- Wallet-to-wallet transfers between individual partner sellers
- Reward balance expiration — point/currency expiry policies are a Phase 2 decision; balances do not expire in v1

6. Users and Personas

6.1 Vendor Personas

Persona	Primary Needs
Client Admin	Configure which redemption options are available for their tenant. Set minimum thresholds and processing modes. View and export full tenant redemption history. Approve or reject return requests.
TenXEngage Platform Admin	Manage the global redemption catalog (XTRM and Xoxoday items). Enable or disable catalog items globally. Manage vendor API credentials and integration health.
Approver	Review and approve or reject pending redemption requests (APPROVAL_REQUIRED mode) and return requests submitted by partner users.

6.2 Partner Personas

Persona	Primary Needs
Partner Seller / Account Executive	Browse the redemption catalog filtered to their available currency balances. Redeem individual wallet balance for cash or non-cash rewards. View transaction history and submit return requests.
Partner Admin	Redeem from the company wallet on behalf of the partner organization. Monitor company wallet balance and transaction history. Export company redemption data.
Partner SE / Solutions Architect	Redeem non-cash rewards (events, swag, experiences) earned through technical enablement and community contribution.

7. Core Use Cases

- Browse a currency-aware non-cash catalog showing only items redeemable with the user's held currency types (points, credits, tickets)
- Redeem from a personal wallet for cash payout via XTRM (ACH, PayPal, Venmo, international transfer)
- Redeem from a personal wallet for non-cash reward via Xoxoday (gift card, merchandise, travel, event)
- Redeem from a company wallet as a Partner Admin

- Track redemption request through its full lifecycle: submitted → reserved → processing → completed
- View full transaction history with status tracking and export to CSV or XLSX
- Submit a return request for a completed non-cash redemption within the client-configured return window
- Client Admin approves or rejects a return request; Xoxoday confirms before wallet is credited
- Client Admin configures tenant-specific catalog enablement, processing modes, and minimum thresholds
- TenXEngage receives vendor webhooks from XTRM and Xoxoday and updates transaction status in real-time

8. Functional Scope

8.1 Reward Wallet

Every partner seller has an individual reward wallet and every partner company has a pooled company wallet. Both are scoped to their client (tenant) via `client_id`.

- The wallet maintains a separate running balance per currency type. The four supported types are: cash, points, credits, and tickets. Each type is tracked independently — a user may simultaneously hold balances across multiple types.
- Each currency balance has two components: `availableBalance` (spendable) and `reservedBalance` (locked against in-flight redemptions). The effective spendable amount is always `availableBalance`.
- Company wallet balances are distinct from and do not aggregate individual seller balances.
- A wallet is eligible for redemption only when the `availableBalance` for the relevant currency type meets or exceeds the client-configured minimum balance threshold.

Acceptance Criteria

- Individual and company wallets display correct available and reserved balances per currency type
- Wallet balance updates in real-time when rewards are earned or redemptions are processed
- Users below the minimum balance threshold cannot initiate a redemption

8.2 Ledger Engine

TenXEngage maintains the authoritative reward ledger. All balance movements are recorded as immutable `LedgerEntry` records before wallet totals are updated. Vendors are fulfillment engines — they do not hold or manage balances.

- Reward earning: CREDIT entry, `availableBalance` increases
- Redemption submission: RESERVE entry, amount moves from `availableBalance` to `reservedBalance`
- Fulfillment confirmed (COMPLETED webhook): DEBIT entry, `reservedBalance` decrements permanently
- Fulfillment failed (FAILED/CANCELLED webhook): RELEASE entry, `reservedBalance` returns to `availableBalance`
- Return confirmed by vendor: RETURN_CREDIT entry, `availableBalance` restored in originating currency type

Acceptance Criteria

- Every balance mutation produces an immutable `LedgerEntry` before wallet totals change
- Wallet totals are always derivable from ledger entries — they are maintained as running aggregates for performance
- Failed or cancelled redemptions fully restore the reserved amount to `availableBalance`

8.3 Redemption Catalog

The global catalog is managed by the TenXEngage platform team in partnership with XTRM and Xoxoday. Client Admins configure a tenant-level view from this global catalog.

- Cash catalog items (via XTRM): ACH bank transfer, PayPal, Venmo, international wire/digital transfer
- Non-cash catalog items (via Xoxoday): gift cards, merchandise, prepaid cash cards (Visa/Mastercard), travel experiences, vendor events
- Each catalog item carries: category (CASH/NON_CASH), compatible currency type(s), minimum redemption amount, default processing mode, geographic availability, and provider item ID
- Client Admins can enable or disable any global catalog item for their tenant, and override the processing mode per item
- Non-cash catalog display is currency-aware: users see only items compatible with the currency types they currently hold. A user with points sees points-redeemable items; a user with tickets sees ticket-redeemable items; a user with both sees items for either, organized by currency type
- Items exceeding the user's availableBalance are displayed but marked unavailable with the shortfall shown

Acceptance Criteria

- Catalog display filters correctly by user's held currency types
- Client Admin can enable/disable items and the changes are immediately reflected to tenant users
- Items outside the user's geographic region are not shown

8.4 Redemption Flow

Redemption requests route automatically to the correct vendor based on catalog item category — CASH to XTRM, NON_CASH to Xoxoday. The user interacts with a single unified UI throughout.

Redemption Category	Vendor	Examples
CASH	XTRM	ACH, PayPal, Venmo, international bank transfer
NON_CASH	Xoxoday	Gift cards, merchandise, prepaid Visa/MC, travel, events

Processing modes are configurable per catalog item at the global level and overridable at the tenant level:

- Instant: request sent immediately to vendor; balance debited on COMPLETED webhook
- Batch: requests queued and processed on a client-configured schedule (daily, weekly)
- Approval Required: request held in PENDING_APPROVAL; Client Admin or Approver must approve before vendor handoff

On submission, regardless of processing mode, the redemption amount moves from availableBalance to reservedBalance, preventing double-spend. The vendor receives the request only after any required approval gate has been passed.

Acceptance Criteria

- CASH redemptions route to XTRM; NON_CASH redemptions route to Xoxoday — transparently
- APPROVAL_REQUIRED requests are not sent to the vendor until approved
- Balance is reserved immediately on submission regardless of processing mode
- COMPLETED webhook triggers ledger DEBIT; FAILED/CANCELLED triggers RELEASE

8.5 Returns

Returns are supported for non-cash (Xoxoday) redemptions only. Cash payouts via XTRM are non-returnable once disbursed.

- Eligibility: redemption must be in COMPLETED status, within the client-configured return window, and the catalog item must not be marked non-returnable (e.g., instantly delivered digital gift cards)
- User submits return request from their transaction history with an optional reason — status set to PENDING_APPROVAL
- Client Admin or designated Approver reviews and approves or rejects
- On approval: TenXEngage notifies Xoxoday via API. Wallet credit is not issued until Xoxoday confirms
- On Xoxoday confirmation: RETURN_CREDIT ledger entry written; full original amount restored to availableBalance in the originating currency type
- On Xoxoday rejection (e.g., item already used): status set to RETURN_REJECTED; no wallet credit; user notified
- On Client Admin rejection: Xoxoday is not contacted; status set to RETURN_REJECTED; user notified
- Partial returns are not supported in v1 — full amount only

Return Status Lifecycle:

Status	Description
PENDING_APPROVAL	Return request submitted by user; awaiting Client Admin review
APPROVED	Client Admin approved; return request sent to Xoxoday; awaiting vendor confirmation
RETURN_CONFIRMED	Xoxoday confirmed the return; wallet credit issued
RETURN_REJECTED	Client Admin rejected the request, or Xoxoday declined after approval
CANCELLED	User cancelled their own request before Client Admin review

Acceptance Criteria

- Return option only shown for eligible completed non-cash redemptions within the return window
- Wallet is credited only after Xoxoday confirmation — not on Client Admin approval alone
- RETURN_CREDIT ledger entry is distinguishable from a standard CREDIT (reward earned) entry

8.6 Transaction History and Reporting

- All redemption and return transactions are persisted with: transaction ID, wallet, catalog item, currency type, amount, status, timestamps, provider reference ID, and vendor name (XTRM or XOXODAY)
- Users can filter history by date range, status, and redemption type
- Client Admins can view the full tenant transaction history across all users and companies
- Transaction history is exportable as CSV or XLSX for both individual users and Client Admins
- Transaction statuses use the existing TransactionStatus enum: PENDING, APPROVED, REJECTED, PROCESSING, COMPLETED, FAILED, CANCELLED

Acceptance Criteria

- Transaction history is paginated, filterable, and exportable
- Client Admin export includes all tenant users and companies
- Return transactions are linked to their originating redemption transaction

8.7 Vendor Integration

XTRM (Cash) — API-first global disbursement platform supporting 180+ countries and 60+ currencies. Handles KYC/AML, OFAC screening, and 1099/W-8BEN tax reporting natively. TenXEngage passes user identity fields (name, email, country) at cash redemption time.

Xoxoday (Non-Cash) — Global rewards catalog platform with localized options in 100+ countries. Handles gift card delivery, merchandise fulfillment, and order lifecycle management. Catalog synced periodically to keep TenXEngage RedemptionCatalogItem table current.

- Vendor webhook endpoints are vendor-specific: /api/v1/redemption/webhook/xtrm and /api/v1/redemption/webhook/xoxoday
- Both webhook streams normalize into the shared TransactionStatus enum and RedemptionTransaction model
- Webhook authenticity verified via HMAC-SHA256 signature using vendor-specific signing secrets stored as encrypted platform configuration
- No payment credentials, bank account numbers, or card details are stored in TenXEngage systems

Acceptance Criteria

- Webhook events from both vendors correctly update transaction and wallet state
- Invalid or unauthenticated webhook requests are rejected with 401
- Vendor API credentials are never logged or exposed in error messages

8.8 Reward Balance Expiration

Reward balances do not expire in v1. This is an explicit product decision — balances accumulate indefinitely until redeemed or manually adjusted. Expiration policy is deferred to Phase 2.

- When expiration is introduced in Phase 2, it will be configurable per client and per currency type (e.g., points expire after 12 months of inactivity; cash does not expire)
- Expiration events will write a EXPIRY_DEBIT LedgerEntry and trigger a notification to the affected user before and at the point of expiry
- Expired balances will be separately reportable so Client Admins can track program liability and breakage rates

Phase 2 Decision

Reward expiration (breakage) is a significant financial and UX decision. The policy — whether tied to inactivity, calendar date, or program end — must be explicitly configured per client and communicated clearly to partners before implementation.

8.9 Redemption Rate and Analytics Dashboard

Redemption rate is a primary health metric for any rewards program — it measures the percentage of earned balances that are actually redeemed. A low redemption rate signals that rewards are not perceived as valuable, thresholds are too high, or the catalog is misaligned with partner preferences.

The following redemption analytics must be available to Client Admins:

- Redemption rate by currency type: total redeemed / total earned, expressed as a percentage
- Redemption rate by catalog item: which items are most and least redeemed
- Redemption rate by partner tier, region, and role: surface which segments are engaging vs. not
- Average time-to-first-redemption: how long after earning do partners redeem for the first time
- Unredeemed balance report: total outstanding liability by currency type across all wallets
- Failed and cancelled redemption rate: to identify fulfillment friction

Acceptance Criteria

- Client Admin dashboard displays redemption rate per currency type with trend over time
- Unredeemed balance report is exportable
- Analytics filter by partner tier, region, and date range

8.10 Payout Speed and SLAs

Partners expect timely reward delivery. Delayed payouts reduce the motivational effectiveness of the program. Processing mode SLAs must be defined and surfaced to users at the point of redemption.

Processing Mode	Expected Payout Timeline	User Communication
Instant	Cash (XTRM): within 1–2 business days depending on payment method. Non-cash (Xoxoday): digital delivery within minutes; physical items per Xoxoday fulfillment SLA.	Confirmation shown immediately on submission with estimated delivery timeframe
Batch	Processed on client-configured schedule (daily or weekly). Payout follows vendor SLA after batch runs.	User shown next batch date at submission. Notification sent when batch is processed.
Approval Required	Clock starts after Client Admin approves. Vendor SLA applies from approval timestamp.	User notified of pending approval status. Second notification when approved and processing begins.

- Expected delivery timelines must be shown to users at the point of catalog item selection — before they submit the redemption
- SLA breach monitoring is a Phase 2 operations requirement — flagging redemptions that exceed expected timelines

Acceptance Criteria

- Estimated delivery timeframe is displayed on the catalog item detail and redemption confirmation screen
- Batch mode shows the next scheduled processing date at submission time

8.11 Persistent Balance Visibility

Industry research consistently identifies persistent balance display as one of the highest-valued features in redemption platforms. Partners should always know what they have available — not just when they visit the redemption store.

- The user's available wallet balance per currency type must be visible in the platform navigation (e.g., header or sidebar) on every page — not only within the Redemption Store
- The balance display shows availableBalance only (not reservedBalance) to avoid confusion
- Clicking the balance widget navigates directly to the Redemption Store
- Balance updates in real-time when rewards are earned or a redemption is completed
- For users with multiple currency balances, display the most contextually relevant currency (e.g., points on an enablement page, cash on a deals page) with a tooltip to expand all balances

Acceptance Criteria

- Wallet balance is visible in the platform header or nav on every authenticated page
- Balance reflects availableBalance in real-time — reservedBalance is not included
- Clicking the balance widget navigates to the Redemption Store

8.12 Regional Catalog Configuration

Global partner ecosystems have diverse reward preferences by region. A catalog item that is highly valued in North America (e.g., an Amazon gift card) may have low relevance in APAC or EMEA. Client Admins must be able to configure region-specific catalog subsets within their tenant.

- Client Admins can configure catalog item enablement at the region level within their tenant — enabling an item for APAC partners but not EMEA, for example
- Region is derived from the user's partner company region attribute, which maps to Xoxoday's geographic availability data
- Regional configuration is layered on top of global catalog availability — a Client Admin cannot enable an item for a region where Xoxoday does not support it
- If no regional override is set, the tenant-level default enablement applies to all regions
- Non-cash catalog items from Xoxoday will surface localized options by default (e.g., local gift card brands in the user's country) based on Xoxoday's region-aware catalog API

Acceptance Criteria

- Client Admin can configure catalog enablement per region within their tenant
- Partner sellers in different regions of the same tenant see correctly filtered catalogs
- Xoxoday's region-aware catalog API drives localized item display within enabled categories

9. Key Differentiators

Differentiator	Description
Unified Redemption Experience	Users interact with a single Redemption Store — XTRM and Xoxoday routing is completely transparent. No vendor switching, no separate portals.
Currency-Aware Catalog	Non-cash catalog surfaces only items redeemable with the currencies the user actually holds. No irrelevant options, no confusion.
Ledger-First Architecture	TenXEngage owns the authoritative ledger. Immutable LedgerEntry records are written before any wallet total changes — ensuring correctness independent of vendor uptime.
Dual Wallet Model	Individual seller wallets and company pooled wallets operate independently, supporting both personal and team-level incentive structures.
Two-Step Balance Safety	Reserve-on-submission, debit-on-confirmation prevents double-spend without blocking the user experience during async fulfillment.
Returns with Vendor Gate	Non-cash returns require Xoxoday confirmation before the wallet is credited — preventing fraud while keeping the experience admin-controlled.
Global Cash + Local Non-Cash	XTRM covers 180+ countries for cash; Xoxoday covers 100+ countries with localized non-cash catalogs. Full global partner coverage.
Compliance Delegated	KYC/AML, OFAC screening, and tax reporting for cash payouts are fully handled by XTRM — zero compliance build required inside tenXengage.
Persistent Balance Visibility	Wallet balance visible platform-wide in the nav on every page — not buried in a redemption store. Partners always know what they have earned.
Redemption Rate Analytics	Client Admins see earned vs. redeemed ratios by currency type, catalog item, tier, and region — the primary health metric for any rewards program.
Regional Catalog Control	Client Admins can configure region-specific catalog subsets. Xoxoday surfaces localized reward options by country automatically.

10. Permissions and Feature Flags

10.1 Feature Flag

Feature Flag Key	Starter	Professional	Enterprise
redemption_store	Enabled	Enabled	Enabled

The redemption_store feature flag can be overridden at the tenant level by the TenXEngage platform team (e.g., disabled during onboarding). All three tier booleans are enabled from launch.

10.2 Permission Keys

Permission Key	Description	Default Roles
module.redemption_store	Access to the Redemption Store module	Partner Seller, Partner Admin, Client Admin
action.redemption.redeem	Initiate redemption from personal wallet	Partner Seller
action.redemption.redeem_company	Initiate redemption from company wallet	Partner Admin
action.redemption.view_history	View own redemption transaction history	Partner Seller, Partner Admin
action.redemption.view_all_history	View all tenant redemption history	Client Admin
action.redemption.export	Export redemption transaction data	Partner Seller, Partner Admin, Client Admin
action.redemption.configure	Configure tenant catalog and thresholds	Client Admin
action.redemption.approve	Approve or reject pending redemption requests	Client Admin, Approver
action.redemption.return.request	Submit a return request for a completed non-cash redemption	Partner Seller, Partner Admin
action.redemption.return.review	Approve or reject return requests	Client Admin, Approver
action.redemption.catalog.manage	Manage the global redemption catalog	TenXEngage Platform Admin

11. Data Model

11.1 Core Entities

- RewardWallet · LedgerEntry
- RedemptionCatalogItem · ClientRedemptionConfig
- RedemptionTransaction · RedemptionReturn

All entities are tenant-scoped: implement TenantAware, extend BaseEntity, carry client_id FK with Hibernate @Filter, and include a Flyway migration with an index on client_id. All IDs are UUID. Timestamps use Instant.

11.2 RewardWallet

Field	Type	Description
id	UUID (PK)	Primary key
clientId	UUID (FK)	Tenant scoping — references clients table
ownerType	Enum	INDIVIDUAL or COMPANY
userId	UUID (FK, nullable)	Set when ownerType = INDIVIDUAL
partnerCompanyId	UUID (FK, nullable)	Set when ownerType = COMPANY
currencyType	Enum	CASH, POINTS, CREDITS, or TICKETS
availableBalance	BigDecimal	Current spendable balance
reservedBalance	BigDecimal	Amount locked against in-flight redemptions
totalEarned	BigDecimal	Lifetime total earned (for reporting)

11.3 LedgerEntry

Field	Type	Description
id	UUID (PK)	Primary key
clientId	UUID (FK)	Tenant scoping
walletId	UUID (FK)	The wallet this entry belongs to
entryType	Enum	CREDIT, DEBIT, RESERVE, RELEASE, RETURN_CREDIT
sourceType	Enum	REWARD_EARNED, REDEMPTION_RESERVE, REDEMPTION_DEBIT, REDEMPTION_RELEASE, RETURN_CREDIT, ADJUSTMENT, CLAWBACK
currencyType	Enum	CASH, POINTS, CREDITS, or TICKETS
amount	BigDecimal	Absolute value of the movement (always positive)
relatedTransactionId	UUID (nullable)	References RedemptionTransaction if from a redemption
createdAt	Instant	Immutable — inherited from BaseEntity

11.4 RedemptionCatalogItem

Field	Type	Description
id	UUID (PK)	Primary key
name	String	Display name
category	Enum	CASH or NON_CASH
subCategory	String	ACH, GIFT_CARD, MERCHANDISE, TRAVEL, EVENT, etc.
currencyType	Enum	Compatible currency type
minimumAmount	BigDecimal	Minimum redemption amount

Field	Type	Description
defaultProcessingMode	Enum	INSTANT, BATCH, or APPROVAL_REQUIRED
providerItemId	String	Reference ID in XTRM or Xoxoday
geographicScope	String[]	ISO country codes where item is available
isReturnable	Boolean	Whether item supports return requests
isActive	Boolean	Whether item is globally active

11.5 ClientRedemptionConfig

Field	Type	Description
id	UUID (PK)	Primary key
clientId	UUID (FK)	Tenant this config belongs to
catalogItemId	UUID (FK)	References RedemptionCatalogItem
isEnabled	Boolean	Whether item is enabled for this tenant
processingModeOverride	Enum (nullable)	Tenant override for processing mode
minimumWalletBalance	BigDecimal	Minimum wallet balance to unlock redemption
minimumTransactionAmount	BigDecimal	Minimum per-transaction redemption amount
returnWindowDays	Integer	Days after fulfillment within which returns are accepted

11.6 RedemptionTransaction

Field	Type	Description
id	UUID (PK)	Primary key
clientId	UUID (FK)	Tenant scoping
walletId	UUID (FK)	Source wallet
catalogItemId	UUID (FK)	Item being redeemed
requestedBy	UUID (FK)	User who initiated the request
amount	BigDecimal	Amount redeemed
currencyType	Enum	Currency type of the redemption
status	Enum	TransactionStatus (PENDING, PROCESSING, COMPLETED, etc.)
processingMode	Enum	INSTANT, BATCH, or APPROVAL_REQUIRED at time of request
providerName	Enum	XTRM or XOXODAY — set at routing time
providerTransactionId	String (nullable)	Reference ID from XTRM or Xoxoday
approvedBy	UUID (nullable)	Approver user ID (APPROVAL_REQUIRED mode)
approvedAt	Instant (nullable)	Timestamp of approval

Field	Type	Description
completedAt	Instant (nullable)	Timestamp of provider confirmation
failureReason	String (nullable)	Provider failure message if FAILED
returnId	UUID (nullable)	References RedemptionReturn if a return was filed

11.7 RedemptionReturn

Field	Type	Description
id	UUID (PK)	Primary key
clientId	UUID (FK)	Tenant scoping
redemptionTransactionId	UUID (FK)	The completed redemption being returned
requestedBy	UUID (FK)	User who submitted the return request
status	Enum	PENDING_APPROVAL, APPROVED, RETURN_CONFIRMED, RETURN_REJECTED, CANCELLED
returnReason	String (nullable)	Optional reason from the user
reviewedBy	UUID (nullable)	Client Admin or Approver who reviewed
reviewedAt	Instant (nullable)	Timestamp of admin review
providerReturnId	String (nullable)	Xoxoday return reference ID
providerConfirmedAt	Instant (nullable)	Timestamp of Xoxoday confirmation
walletCreditedAt	Instant (nullable)	Timestamp when wallet credit was issued
rejectionReason	String (nullable)	Reason for rejection (admin or vendor)

11.8 Key Relationships

- RewardWallet has one balance record per currency type per owner (user or company)
- LedgerEntry belongs to RewardWallet; every balance mutation produces one LedgerEntry
- RedemptionTransaction belongs to RewardWallet and RedemptionCatalogItem; references optional RedemptionReturn
- RedemptionReturn belongs to RedemptionTransaction; one return per transaction maximum
- ClientRedemptionConfig belongs to Client and RedemptionCatalogItem; defines the tenant-level view

12. API Surface

All endpoints follow platform conventions: /api/v1/ base path, kebab-case segments, UUID IDs, camelCase JSON. All endpoints require Authorization: Bearer <jwt-token> and X-Client-Subdomain headers.

12.1 Wallet APIs

Method	Path	Permission	Description
GET	/api/v1/wallets/me	module.redemption_store	Get current user's wallet balances per currency type
GET	/api/v1/wallets/company/{companyId}	action.redemption.redemption_company	Get company wallet balances per currency type

12.2 Catalog APIs

Method	Path	Permission	Description
GET	/api/v1/redemption/catalog	module.redemption_store	Get tenant-enabled catalog items, filtered by user's held currency types
GET	/api/v1/redemption/config	action.redemption.configure	Get tenant redemption configuration
PUT	/api/v1/redemption/config	action.redemption.configure	Update tenant redemption configuration

12.3 Redemption APIs

Method	Path	Permission	Description
POST	/api/v1/redemption/requests	action.redemption.redemption	Submit redemption from personal wallet
POST	/api/v1/redemption/requests/company	action.redemption.redemption_company	Submit redemption from company wallet
GET	/api/v1/redemption/requests	action.redemption.view_history	List own redemption transactions (paginated)
GET	/api/v1/redemption/requests/{id}	action.redemption.view_history	Get specific transaction detail
GET	/api/v1/redemption/requests/all	action.redemption.view_all_history	List all tenant transactions (Client Admin)
POST	/api/v1/redemption/requests/{id}/approve	action.redemption.approve	Approve a pending redemption request
POST	/api/v1/redemption/requests/{id}/reject	action.redemption.approve	Reject a pending redemption request
GET	/api/v1/redemption/requests/export	action.redemption.export	Export transactions as CSV or XLSX

12.4 Returns APIs

Method	Path	Permission	Description
POST	/api/v1/redemption/requests/{id}/returns	action.redemption.return.request	Submit a return request for a completed non-cash redemption
GET	/api/v1/redemption/returns	action.redemption.return.request	List own return requests (paginated)

Method	Path	Permission	Description
POST	/api/v1/redemption/returns/{id}/approve	action.redemption.return.review	Approve a return request (Client Admin)
POST	/api/v1/redemption/returns/{id}/reject	action.redemption.return.review	Reject a return request (Client Admin)
DELETE	/api/v1/redemption/returns/{id}	action.redemption.return.request	Cancel own pending return request

12.5 Webhook APIs

Method	Path	Auth	Description
POST	/api/v1/redemption/webhook/xtrm	HMAC-SHA256	Inbound webhook from XTRM for cash payout status updates
POST	/api/v1/redemption/webhook/xoxoday	HMAC-SHA256	Inbound webhook from Xoxoday for non-cash order status updates

13. Event Architecture

The Redemption Store emits platform events for downstream integration with notifications, analytics, audit, the PAS scoring pipeline, and future AI services:

- redemption_requested · redemption_approved · redemption_rejected · redemption_processing
- redemption_completed · redemption_failed · redemption_cancelled
- return_requested · return_approved · return_confirmed · return_rejected · return_cancelled
- wallet_balance_updated · wallet_below_minimum_threshold
- vendor_webhook_received · vendor_webhook_failed

PAS Integration	Redemption events are candidate signals for the Partner Activation Score Commercial Intent component (18% weight). Reward redemption activity indicates active program participation and can contribute to PAS scoring in Phase 2.
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14. Non-Functional Requirements

Category	Requirement
Security	Tenant isolation enforced via client_id, Hibernate @Filter, and TenantValidator. No payment credentials stored in tenXengage. All vendor communication over TLS 1.2+. Webhook authenticity via HMAC-SHA256.
Ledger Integrity	All balance mutations are idempotent and transactional. LedgerEntry records are immutable once written. Wallet totals must always be consistent with the ledger sum.

Category	Requirement
Performance	Wallet balance reads and catalog browsing must be responsive under normal tenant scale. Vendor webhook processing must handle concurrent events without race conditions on wallet totals.
Reliability	Redemption submission, ledger writes, and webhook processing must be idempotent. Background job retries for vendor API calls on transient failure. Dead-letter handling for unprocessable webhooks.
Compliance	KYC/AML, OFAC screening, and tax reporting for cash payouts fully delegated to XTRM. No PII stored beyond what is necessary for vendor handoff. Anti-bribery compliance module hooks maintained.
Observability	Structured logging with MDC (requestId, clientId, userId) on all redemption flows. Metrics for redemption volume, vendor response time, webhook processing latency, and failure rate.
Multi-Tenant Config	All thresholds, processing modes, catalog enablement, and return windows are fully tenant-configurable. No hardcoded values per tenant.
Accessibility	Redemption Store UI follows WCAG-aligned patterns consistent with the broader tenXengage platform.

15. UX Principles

- The Redemption Store feels like a seamless part of tenXengage — not a third-party portal. No vendor branding or names visible to end users.
- Currency-aware catalog display eliminates irrelevant options. Users only see what they can actually redeem with their current balance.
- Balance state is always clear — available vs. reserved is surfaced at a glance, with shortfalls shown inline on unavailable items.
- Transaction history is the user's receipt book — always accessible, always filterable, always exportable.
- Return flows are straightforward: find the transaction, request the return, track the status. No confusion about what happens next.
- Admin configuration is self-serve and low-friction — Client Admins can manage their catalog and thresholds without platform team involvement.
- Wallet balance is always visible — displayed in the platform header or nav on every page so partners never have to hunt for their balance.
- Payout timelines are set upfront — expected delivery time is shown at the catalog item level before the user submits, not after.
- Regional relevance by default — Xoxoday catalog surfaces locally relevant options first, reducing irrelevant choices for global partner ecosystems.

16. Release Strategy

Phase 1 — Production v1

Track	Scope
Track A: Must Ship	Wallet model (individual + company) · Ledger engine (CREDIT, RESERVE, DEBIT, RELEASE) · Redemption catalog (global + client config) · XTRM cash payout integration · Xoxoday non-cash integration · Vendor webhook

Track	Scope
	processing · Transaction history and export · Permissions and feature flags · Notifications · Audit logging
Track B: Differentiated	Returns flow (non-cash only) · Currency-aware catalog filtering · Return window configuration · Xoxoday return confirmation flow · RETURN_CREDIT ledger entry
Track C: Defer	PAS signal integration for Commercial Intent · Batch processing scheduler UI · Advanced redemption analytics · Mobile-native redemption experience

Phase 2 — Enhanced Capabilities (3–6 weeks)

- PAS Commercial Intent signal integration — redemption activity contributes to partner scoring
- Batch processing scheduler — configurable cadence per client with UI controls
- Advanced redemption analytics — redemption rate by catalog item, currency type, partner tier, and region
- Partial return support — configurable percentage-based returns
- Reward balance expiration — configurable per client and currency type; expiry notifications; breakage reporting
- Advanced redemption analytics — redemption rate trends, time-to-first-redemption, unredeemed balance liability report
- SLA breach monitoring — flag redemptions exceeding expected payout timelines
- Wallet statement view — full ledger history exportable per user

Phase 3 — Advanced Platform

- Real-time redemption analytics with cohort comparisons (redeemers vs. non-redeemers on program performance)
- AI-driven redemption nudges — recommending redemption options based on balance thresholds and partner behavior
- Native mobile redemption experience
- Cross-currency redemption options — configurable conversion rules

17. User Stories

Partner Seller Stories

- As a partner seller, I want to browse the redemption catalog showing only items I can redeem with my points balance, so that I can quickly find relevant options
- As a partner seller, I want to redeem my points for a gift card and receive it without leaving the tenXengage platform
- As a partner seller, I want to redeem my cash balance via PayPal so that I can receive a direct payout for deals I have closed
- As a partner seller, I want to see my full transaction history with status tracking so that I always know where my redemptions stand
- As a partner seller, I want to submit a return request for a merchandise item that was incorrect, so that my balance is restored

Partner Admin Stories

- As a partner admin, I want to redeem from the company wallet to claim a team reward trip that our organization earned

- As a partner admin, I want to view the company wallet balance and transaction history so I can report on our team's redemption activity
- As a partner admin, I want to submit a return on behalf of the company wallet when a non-cash reward cannot be used

Client Admin Stories

- As a client admin, I want to enable only the redemption options that make sense for my partner program, so my partners see a curated, relevant catalog
- As a client admin, I want to set a minimum wallet balance threshold so that partners only redeem once they have earned meaningful rewards
- As a client admin, I want to configure a 30-day return window so that partners have time to flag issues with fulfilled items
- As a client admin, I want to approve or reject return requests from my partners and see the outcome in the transaction history
- As a client admin, I want to export the full tenant redemption history to a spreadsheet for program reporting

18. Acceptance Criteria Summary for v1 Launch

The Redemption Store is ready for v1 launch when:

- Individual and company wallets display correct per-currency-type balances in real-time
- Every balance mutation produces an immutable LedgerEntry before wallet totals are updated
- Catalog displays correctly filtered by user's held currency types and client-enabled items
- Cash redemptions route to XTRM and non-cash redemptions route to Xoxoday — transparently
- Balance is reserved on submission and finalized on COMPLETED webhook; released on FAILED or CANCELLED
- APPROVAL_REQUIRED redemptions are not sent to the vendor until approved by Client Admin or Approver
- Vendor webhooks from both XTRM and Xoxoday are authenticated, processed, and update transaction status correctly
- Transaction history is accessible, filterable, and exportable for users and Client Admins
- Return requests can be submitted, reviewed, and resolved — wallet credit only issued after Xoxoday confirmation
- All redemption and return operations are permission-gated and audited
- Wallet balance is persistently visible in the platform nav on every authenticated page
- Estimated payout timelines are displayed at catalog item selection and on redemption confirmation
- Client Admin analytics dashboard shows redemption rate by currency type
- Regional catalog configuration is respected — partners in different regions see correctly filtered catalogs
- Notifications are sent for all key redemption and return lifecycle events
- APIs and UI are stable for real customer usage

19. Notifications

All notifications are delivered through the platform's existing notification framework under NotificationCategory.REWARDS.

Event	Recipient	Category
Redemption request submitted	Requesting user	REWARDS
Redemption approved (APPROVAL_REQUIRED mode)	Requesting user	REWARDS
Redemption rejected	Requesting user	REWARDS
Redemption completed — vendor confirmed	Requesting user	REWARDS
Redemption failed	Requesting user	REWARDS
Pending redemption awaiting approval	Client Admin / Approver	REWARDS
Return request submitted	Client Admin / Approver	REWARDS
Return approved by Client Admin	Requesting user	REWARDS
Return confirmed by Xoxoday — wallet credited	Requesting user	REWARDS
Return rejected (admin or vendor)	Requesting user	REWARDS
Wallet balance below minimum threshold	Partner Seller / Partner Admin	REWARDS

20. Audit Trail

All operations are logged via the platform's @Audited annotation, aligned with existing AuditAction and AuditResourceType enums.

Operation	AuditAction	AuditResourceType	Actor
Redemption request submitted	CREATED	CLAIM	USER
Redemption approved	APPROVED	CLAIM	USER
Redemption rejected	REJECTED	CLAIM	USER
Tenant catalog configured	EDITED	CLIENT	USER
Return request submitted	CREATED	CLAIM	USER
Return approved by Client Admin	APPROVED	CLAIM	USER
Return rejected	REJECTED	CLAIM	USER
Wallet credited after vendor confirmation	EDITED	CLAIM	SYSTEM
Provider webhook received and processed	SYNCED	DATA	SYSTEM

21. Risks

Risk	Mitigation
Vendor API instability (XTRM or Xoxoday downtime)	Idempotent retry logic with exponential backoff. Redemption requests remain in PENDING/PROCESSING state during outage. Webhook dead-letter queue for unprocessable events.
Ledger inconsistency from concurrent requests	Optimistic locking on wallet balance fields. All ledger writes within a single database transaction. Wallet totals derived from ledger entries if inconsistency is detected.
Webhook replay or spoofing attacks	HMAC-SHA256 signature verification on all inbound webhooks. Idempotency key checks prevent duplicate processing.
KYC/AML failure for international cash payouts	XTRM handles KYC natively. TenXEngage surfaces clear failure messaging when XTRM rejects a payout due to compliance. Payout reserved balance is released on failure.
Return abuse or fraud (users returning used items)	Xoxoday confirmation required before wallet credit. Client Admin approval gate. isReturnable flag per catalog item to exclude non-returnable items.
Catalog drift between Xoxoday and TenXEngage	Periodic catalog sync job keeps RedemptionCatalogItem table current. Items removed by Xoxoday are auto-deactivated.
Scope creep from analytics and mobile requests	Strict phase gating. Phase 2 and 3 items explicitly deferred in release strategy.
Low redemption rate at launch	Ensure catalog is curated and regionally relevant from day one. Surface balance prominently on every page. Set minimum thresholds conservatively — they can be raised later.
Regional catalog misconfiguration	Xoxoday geographic availability data drives hard limits. Client Admins cannot enable items outside Xoxoday coverage. UI validation enforces this at configuration time.
User confusion over payout timelines	SLA information shown proactively at catalog item selection — before submission. Batch mode shows next run date explicitly.

22. Open Questions

#	Question	Owner	Status
1	Provider selection confirmed: XTRM for cash payouts and Xoxoday for non-cash rewards.	Product / Engineering	CLOSED
2	Should the company wallet redemption require a minimum number of approvers, similar to incentive approval workflows?	Product	Open
3	What is the batch processing cadence for BATCH mode — fixed platform schedule or configurable per client?	Product	Open
4	Are there limits on simultaneous in-flight (reserved) redemption requests per user?	Product / Engineering	Open
5	Should non-cash redemptions support a quantity-based model, or is it always amount-based?	Product	Open
6	How are failed redemptions surfaced to Client Admins — notification only, or a dedicated failed-transaction view?	Product / UX	Open

#	Question	Owner	Status
7	TenXEngage ledger balances in four platform currency types. USD is the base for cash. FX conversion handled by XTRM.	Engineering	CLOSED

23. Recommended Backend Modules

- reward_wallet — wallet model, per-currency balance management, balance eligibility checks
- ledger_engine — LedgerEntry persistence, balance mutation logic, idempotency enforcement
- redemption_catalog — global catalog management, client configuration, currency-aware catalog API
- redemption_orchestration — request submission, vendor routing, processing mode handling, balance reservation
- xtrm_integration — XTRM API client, cash payout submission, webhook handler
- xoxoday_integration — Xoxoday API client, non-cash order placement, catalog sync, webhook handler, return handoff
- returns_engine — return request lifecycle, Client Admin approval flow, Xoxoday confirmation handling
- redemption_reporting — transaction history, export generation, Client Admin reporting

24. Recommended Front-End Areas

Partner Seller / Partner Admin

- Wallet dashboard — per-currency balance display, available vs. reserved, earning history summary
- Redemption Store — currency-aware catalog browser, item detail, redemption form, submission confirmation
- Transaction history — filterable, status-tracked list with export action
- Return request flow — initiation from transaction history, status tracking, outcome notification

Client Admin

- Redemption catalog configurator — enable/disable items, set processing mode overrides, configure thresholds and return windows
- Approval queue — pending redemption approvals and return request reviews
- Tenant reporting — full transaction history with filters and export

Platform Admin

- Global catalog manager — add/edit/deactivate catalog items, manage XTRM and Xoxoday item mappings
- Vendor integration health — API status, webhook delivery logs, catalog sync status

25. Final Recommendation

Proceed with a two-step commitment:

- Deliver a production-ready v1 with the full earn-to-redeem loop: wallet engine, ledger, XTRM cash integration, Xoxoday non-cash integration, currency-aware catalog, transaction history, and returns — built on the platform's existing multi-tenant, permission-gated, audit-logged infrastructure
- Position batch processing analytics, PAS signal integration, partial returns, and mobile redemption as the 3–6 week Phase 2 roadmap

The combination of a ledger-first dual-wallet model, two best-of-breed vendor integrations (XTRM + Xoxoday), and a fully tenant-configurable catalog closes the most important gap in the current platform — the missing earn-to-redeem loop — and does so without tenXengage needing to build fulfillment, compliance, or catalog infrastructure.

Strategic
Note

Redemption activity is a candidate signal for the Partner Activation Score Commercial Intent component (18% weight). As Phase 2 integrates PAS scoring, redemption events should be routed into the ActivationSignal pipeline — connecting every reward redemption to measurable partner revenue readiness.